



No hard ceilings

A rise in minimum PF pension will protect the worse-off among the retired

The Union Cabinet’s approval to raise the wage ceiling for mandatory coverage under the Employees’ Provident Fund Organisation (EPFO) from ₹15,000 to ₹25,000, though belated, is commendable. The BJP-led government has finally responded to this long-standing demand of workers in the organised sector. Around 51 lakh additional employees may be brought under the PF fold, which has around 7.98 crore contributing members. Those in the wage slab of ₹15,000-₹25,000 will now be covered under the Employees’ Pension Scheme (EPS), 2026. Although Union Labour Minister Mansukh Mandaviya cited a government survey as the basis for the increase, the 12-year delay in implementing it should have been avoided. The Minister, who said that the average salary in private establishments was ₹23,000, should also make the survey’s findings public for a better understanding of the country’s overall wage situation. With some concerns having been raised that not every employer may be willing to make the additional contributions required after the revision, the government must ensure that all employers comply with the law.

So far, the government has been silent on revising the EPS minimum pension of ₹1,000, fixed in September 2014 alongside the previous revision in the PF wage ceiling. EPFO pensioners have been demanding the hike in the minimum pension amount. Before it assumed office at the Centre in 2014, the BJP had urged the Congress-led UPA government to keep the minimum pension at ₹3,000 and index it to inflation. Yet, it has not made even a symbolic beginning towards the indexation of pension to inflation. Moreover, about 45% of the nearly 82 lakh pensioners receive ₹1,000 or less. In March 2026, the Joint Parliamentary Standing Committee on Labour reminded the government that “the existing minimum pension amount is inadequate to meet even the basic needs of pensioners particularly in the ... scenario marked by ... rising healthcare and living expenses”. Also, little is known about the follow-up to the panel’s recommendation for “an urgent and comprehensive review of the minimum pension”. The EPFO must release the findings of actuarial assessment, made by valuers in recent years. This should also be beneficial to the PF body, as EPFO CEO Ramesh Krishnamurthi observed in an interview with *The Hindu* in August that “there is a misconception that the [pension] fund has unlimited resources”, while ruling out the reopening of the window for higher pensions. Even if the government is unwilling to restore the pre-2014 coverage, under which all employees were covered regardless of their wages, it should consider relaxation of the restriction on the coverage. Else, many pensioners will be pushed below the poverty line.

Continental glory

India might get another 100-medal haul from a smaller contingent

Even as Japan, known for efficiency, welcomes its third Asian Games and the first in 32 years, on Saturday, the Aichi-Nagoya Games have already earned the ‘worst ever’ tag because of accommodation and logistical issues. A country that successfully hosted two Olympics – in 1964, within two decades of suffering atomic bombings, and in 2021, behind closed doors amid the unprecedented COVID-19 pandemic – is now strangely struggling to accommodate more than 17,000 athletes and officials from 45 nations in hotels, containers and a cruise ship. Despite these issues, 501 of India’s athletes, including 267 men and 234 women, across 36 disciplines, will try to keep its flag flying high. At the previous Hangzhou Asiad, India had an impressive tally of 106 medals and was placed fourth behind China, Japan and South Korea. There is a significant drop in the number of Indian athletes, from 650-plus in 40 disciplines three years ago, following rationalisation, discouraging a mad rush because of the perks and money associated with the Asiad. Nevertheless, Indian sports authorities expect around 100 medals, which may noticeably improve the athlete-to-medal ratio.

The presence of 310 debutants in the contingent signals a brighter future for the country, which has spent nearly ₹800 crore on Asian Games preparations in the last cycle. Shooting and athletics, which together field 105 athletes, should continue to be the major contributors to India’s medals tally, followed by archery, boxing, badminton, squash and wrestling. Stars including Manu Bhaker, Mirabai Chanu, Lovlina Borgohain, P.V. Sindhu, Aman Sehrawat, Tajinderpal Singh Toor, Gulveer Singh, Harmanpreet Singh, Smriti Mandhana and Jasprit Bumrah will lead India’s campaign. At 39, squash player Joshna Chinnappa will be the first Indian to compete in seven consecutive Asian Games. The youngest athlete, 15-year-old cricketer Vaibhav Sooryavanshi, and the oldest, 55-year-old rider Shruti Vora, are the other attractions. Some attention will also be devoted to Indian men’s and women’s hockey teams, which will aim to win gold medals and secure berths at the 2028 Los Angeles Olympics. While India can take pride in fielding athletes in debut-making disciplines, such as mixed martial arts and teqball, it should ponder over its poor representation in swimming, gymnastics and cycling, which offer more medals and open avenues for Olympic glory. Meanwhile, the Indian men and women’s teams would be watched closely in cricket. In pushing for continental excellence, India should also remain vigilant about its deteriorating global reputation due to the sharply rising doping graph.

Technologies that once arrived in India through imports are increasingly being invented at home through local research institutions, industry, and startups. Three institutional foundations of an innovation economy are now advancing together: public research, corporate research and development (R&D), and deep-tech entrepreneurship. The transformation is yet incomplete, but the architecture is finally visible. Gallium Nitride (GaN) semiconductor technologies, critical for advanced radar, space systems, and next-generation communications, are increasingly being developed domestically, helping close a strategic gap in a tightly regulated, export-controlled field. At the same time, affordable immunotherapies developed in India are expanding access to advanced cancer care and demonstrating the country’s growing capacity for biomedical innovation.

Patent filings are often cited as a ready indicator of innovation. The filing numbers continue to show impressive growth. From just over 1,10,000 patent filings in 2024-25, the number grew to more than 1,43,000 in 2025-26, a 30.2% increase. Moreover, domestic applicants now account for almost seven in 10 filings.

However, the sheer volume of applications does not tell the complete story. Even a modest enquiry reveals an innovation story that begins to fray at the edges. The number of patents in force in India (just over 2,40,000 in 2025) is a fraction of those in China (5.7 million), the United States (3.5 million), and Japan (2.1 million); all these are 2024 data.

This suggests low grant rates, high abandonment rates, or a combination of both. Add to this meagre commercialisation, low R&D spending – India still spends just under 1% of GDP on research and development, against about 2.4% in China and 3.5% in the U.S. – and we see how far we are from a truly technology-driven, innovation-led economy.

The emerging pillars

However, there lies a resilient core, and shaping it are three strong and vital strands that promise to anchor India’s future. Public institutions with government support have made vital breakthroughs in long-gestation innovation. Private R&D spending exceeded that of all levels of government combined for the first time in 2024. This is expected to climb and account for 55% through 2025-26. Domestic industrial houses are beginning to reap rewards from focused and committed R&D. A maturing startup ecosystem is finding its feet beyond low-hanging service platforms, actively showcasing deep tech and complex manufacturing. That all three strands



Mukundan Chakrapani

Holds a PhD in science and a law degree and is an academic



Gaurav Jain

Founder of Hastin Research and an intellectual property rights expert helping enterprises, startups, and universities transform ideas into intellectual property

Homegrown research and deep-tech enterprise are helping India move beyond its identity as a services provider

are coming of age together bodes well for India.

In March 2023, Defence Research and Development Organisation (DRDO) scientists at the Solid State Physics Laboratory (SSPL) in Delhi and the Gallium Arsenide Enabling Technology Centre (GAETEC) in Hyderabad announced a breakthrough in making gallium nitride (GaN) monolithic microwave integrated circuits (MMICs). The technical know-how to make these circuits, which are critical for military and defence applications, was, by widely reported accounts, refused to India under the offset provisions of the Rafale fighter jet purchase from France. The DRDO is now actively transferring the technology for GaN High Electron Mobility Transistors (HEMTs)-based MMICs to be used in 5G/6G wireless infrastructure, electric vehicles on-board chargers, and renewable energy inverter systems. India is now one of seven countries to have mastered this technology, alongside China, France, Germany, Russia, South Korea and the U.S. AGNIT Semiconductors, a spin-off from the Centre for Nano Science and Engineering (CeNSE) at the Indian Institute of Science, Bengaluru is translating homegrown GaN technology to real-world commercial applications.

From implementer to innovator

While long-gestation R&D at public institutions is starting to pay off, there is growing optimism from academia and domestic industry-led R&D as well. The Bharat 6G Alliance (B6GA) is making rapid progress in its stated aim of contributing 10% of the global 6G patents by 2030. According to a review meeting by Union Minister for Communications Jyotiraditya Scindia, B6GA members have made more than 7,700 patent filings across 5G and 6G technologies, including over 4,400 foreign filings. But, these are applications, not yet grants or declared standard-essential patents, and their true impact remains to be seen. However, taken together with almost 3,000 technical contributions to 3GPP last year, a 15-fold increase from 2020, these figures point in the right direction. The World Intellectual Property Organization’s 2025 Patent Cooperation Treaty rankings showed that Jio Platforms Limited had broken into the top 20 international patent filers for the first time, rising more than 300 places to be placed 19th overall.

While patent numbers are only a part of the story, participation in standards bodies and intellectual property ownership increasingly determine who captures value from future communication networks. India is securing ownership of critical digital infrastructure and is moving from being a standard-implementer to a standard-setting innovator.

Deep-sea discovery, environmental responsibility

India is standing at the threshold of a new frontier. But the most important question is not how deep we can go, how much we can mine, or how much wealth lies beneath the ocean floor. It is more fundamental: Should everything that can be technologically exploited necessarily be exploited? The question is particularly important as India strengthens its position in deep-sea exploration and develops the scientific and technological capabilities to access one of the least understood parts of the planet.

A growing deep-sea footprint

India is not new to seabed mineral exploration and holds three International Seabed Authority (ISA) exploration contracts covering approximately 95,000 square kilometres across the Central Indian Ocean Basin, Central Indian Ridge and Carlsberg Ridge in the international seabed area. Its exploration has identified an estimated 366 million tonnes of polymetallic nodules containing nickel, copper, cobalt and manganese, alongside extensive environmental and biodiversity studies.

Under the Deep Ocean Mission, India is developing deep-sea mining technology, underwater robotics and the MATSYA-6000 human submersible, while the National Institute of Ocean Technology has tested a mining machine at depths of about 5,270 metres. At the same time, biodiversity surveys across 19 seamounts have studied around 1,300 deep-sea organisms, including nearly 23 species reported as new to science. These achievements also impose a profound responsibility to decide whether technological ability should ultimately lead to exploitation.

Technological progress can create a dangerous assumption: if we can exploit something, we should. But the deep ocean challenges this logic. It is a complex and poorly understood ecosystem, and India’s own biodiversity research continues to reveal previously unknown organisms. We cannot confidently predict the consequences of



P. Ragavan

Marine biologist with over a decade of professional experience in the conservation and management of coastal and marine environments

India has an opportunity to become a global leader not only in deep-sea technology but also in deep-sea ecological governance

large-scale seabed disturbance while our understanding of these ecosystems remains incomplete. This is not an argument against science, but for more science before potentially irreversible intervention. Deep-sea minerals may have legitimate strategic importance for India’s renewable-energy, electric-mobility and advanced-manufacturing ambitions. However, strategic importance cannot automatically become ecological permission. Before considering extraction, India should ask whether the minerals are genuinely necessary at the proposed scale, whether alternatives exist, and whether demand can be reduced through recycling, efficiency, substitution and a circular economy. Necessity must be demonstrated, not presumed.

India’s investment in deep-ocean science need not imply a commitment to commercial mining. Exploration can advance knowledge, biodiversity assessment, environmental baselines and technology without leading to exploitation. Mining, however, would physically disturb the seabed. Therefore, adaptive management alone cannot guarantee ecological protection. India has an opportunity to adopt a stronger principle before exploitation begins: the question should not be “How can we mine with minimum damage?” but “Do we need to mine at all?” Since India’s ISA activities remain at the exploration stage, this is the right time to establish a high ecological threshold for any future decision to exploit.

Science must also teach us to say ‘no’

The deep ocean is now emerging as the next frontier for economic growth. Yet, we expect the ocean to regulate climate and support biodiversity while extending industrial activity into its least understood ecosystems. The ocean should not become a place where the environmental costs of human consumption are simply relocated. Scientific progress should not mean making more of nature exploitable and

There is also a marked shift in India’s startup ecosystem. Buoyed by the government’s commitment to Research, Development, and Innovation (RDI) and private capital matched by initiatives such as the more than \$2.5 billion in deep-tech commitments made by members of the India Deep Tech Alliance (IDTA), Indian startups are aiming farther and higher than before. Pixxel Space, founded by two Birla Institute of Technology and Science (BITS) Pilani alumni, is carving a niche in hyperspectral imaging technology with six satellites in orbit and a full constellation of 18 to 24 planned. The Vikram-1 low earth orbit launch by Skyroot Aerospace and the IIT Madras nurtured Agnikul Cosmos’s parallel push towards fully reusable launch vehicles promise to strengthen India’s role as a frugal space technology leader.

Startups such as ImmunoACT, incubated at IIT Bombay in partnership with the Tata Memorial Centre, are expanding access to advanced and affordable health care. NexCARI9, India’s first indigenous CART cell therapy, has delivered effective cancer treatment at a tenth of typical treatment costs. Leveraging the Biotechnology Industry Research Assistance Council (BIRAC) early-stage support, Bengaluru-based Remidio Innovative Solutions is at the forefront in the fight against preventable blindness, such as diabetic retinopathy and glaucoma, through smartphone enabled retinal imaging systems and AI.

The direction is clear

These examples do not by themselves make India an innovation superpower but demonstrate that the three strands are coming together to weave an enduring innovation ecosystem. The GaN story illustrates the emerging innovation pipeline. Government laboratories developed the foundational capability, academic institutions nurtured the talent and research ecosystem, and startups such as AGNIT are now carrying the technology into commercial markets.

However, sustaining these three strands will demand more work, from standardising tech-transfer terms for publicly funded IP and expanding examiner capacity at the patent office, to funding the translational gap between a granted patent and a first paying customer. Yet, as we look through the innovation lens, the direction is clear. India is gradually shedding its identity as a back-office service provider and strengthening the institutions that enable nations to generate their own technology, intellectual property, and industrial capabilities. The journey remains unfinished, but the contours of an innovation economy are beginning to form.

The views expressed are personal

LETTERS TO THE EDITOR

Tariff threats, diplomacy

While maintaining a robust bilateral relationship with the United States is strategically vital, foreign policy cannot be dictated by the threat of economic coercion (Page 1, “U.S. clears Russia sanctions Bill; India faces tariff threat”, September 18). True partnerships are built on mutual respect and an understanding of each other’s domestic

compulsions. Imposing 100% tariffs would not only strain India-U.S. ties but could also destabilise international energy markets, leading to wider global economic friction. India must continue its balanced diplomatic approach — diversifying its energy sources while keeping communication channels open with Washington to resolve trade frictions through dialogue

rather than compliance.
Nagarajamani M.V.,
Hyderabad

Tata’s crisis of governance

Despite the leading roles played by the Tata Group in India’s growth in semiconductors, defence, software and electronics, the recent governance crisis and boardroom divisions at Tata Sons over leadership tenure and regulatory pressures pose a systemic

risk to the broader economy. The possible legal deadlock involving the Tata Trusts and regulatory mandates could stall vital national projects, highlighting the urgent need for an amicable resolution rather than protracted feuds. This is essential both in the national interest and to uphold the reputation of the Tata name, which is held in high esteem by the

people of India.
Rajakumar Arulanandham,
Palayamkottai, Tamil Nadu

There is something quietly damning about a company that discovers the value of “leadership continuity” only when it needs to appear stable to outsiders. What this exposes is how thin the group’s commitment to its own governance standards becomes when a market debut is at stake. An

‘outgoing’ chairman’s stated principle was treated as a formality to be abandoned whenever it no longer suited investor optics. That is not stewardship. It is a company discovering, rather conveniently, that its rules mattered only until they became inconvenient.

Yashvi Patel,
Anand, Gujarat

Letters emailed to letters@thehindu.co.in must carry the postal address.